Pandelis Zembashis

CTO | Hands-On Systems Engineer | AI Developer Infrastructure & Product

p@ndel.is | pa.ndel.is | github.com/PandelisZ | linkedin.com/in/Pandelis

EXECUTIVE SUMMARY

CTO and hands-on systems engineer building AI-native developer infrastructure across CLI, browser, enterprise deployment, desktop, and IDE-oriented surfaces. At Cosine, lead product and engineering for a sovereign AI coding platform selected for the UK Government's £500m Sovereign AI programme and awarded AIRR supercompute backing for British-owned model development. Relevant to xAI London roles across Grok API/developer experience, sandboxed execution, SRE-backed infrastructure, and high-ownership product systems.

PROFESSIONAL EXPERIENCE

Cosine | CTO | Oct. 2023 – Present

Sovereign AI coding product spanning CLI, browser, enterprise deployment, desktop, and IDE-oriented surfaces.

- Own product and engineering for Cosine's sovereign AI coding platform, selected for the UK Government's £500m Sovereign AI programme and awarded 500,000 AIRR GPU hours on Isambard-AI.
- Architected Cosine CLI as a portable single Go binary with browser use, internet research, memories, and bundled ripgrep to reduce installation friction across product surfaces.
- Designed an event-bus agent loop with bidirectional UI events, enabling shared workflows across CLI, desktop, and extension surfaces.
- Built Kubernetes/gVisor cloud-agent execution for secure code execution, tool use, air-gapped deployment, and reliability-focused feedback loops.

11x.ai | Head of Engineering | Apr. 2023 – Sep. 2023

Fastest-growing UK AI startup from 0 to $1M ARR in under six months. TechCrunch coverage.

- Scaled the codebase and engineering organization behind Alice as 11x.ai grew from 0 to $1M ARR in under six months.
- Scaled outbound email and event-driven automation systems that processed millions of email chains and billions of tokens.
- Established onboarding and operating rhythms that kept the engineering team shipping through rapid seed-to-Series A growth.

Papaya | Technical Lead | Feb. 2022 – Mar. 2023

Marketplace and fleet platform helping electrify commercial and last-mile logistics. TechCrunch coverage.

- Led delivery of a fleet management platform for light electric vehicles in Next.js, React Query, and Tailwind CSS.
- Modernized the product API layer by introducing GraphQL into the Ruby on Rails backend, improving iteration across product and engineering.
- Established development cadences and product intake processes that improved execution across engineering and product.

Weezy (acquired by Getir) | Senior Software Engineer | Aug. 2021 – Jan. 2022

Rapid grocery delivery startup building warehouse, ordering, and customer systems.

- Took over the core ordering and processing monolith in Ruby on Rails, owning day-to-day delivery and forward planning during a high-growth phase.
- Built warehouse operations systems and supporting AWS and Heroku infrastructure with a focus on scale, availability, and security.

finleap connect | Senior Software Engineer | Nov. 2020 – Aug. 2021

White-label fintech platform powering digital banking products and apps.

- Built GraphQL APIs on NestJS and cross-platform clients in React Native, Apollo, Node.js, and TypeScript for a multi-tenant fintech platform.
- Led alerting and observability improvements while supporting migration from ECS to Amazon Lambda.
- Owned client-specific delivery across the stack, from GraphQL resolvers and mutations to customer-facing mobile screens.

IoT CONNCTD GmbH | Full Stack Engineer | Jul. 2019 – Nov. 2020
Cloud IoT platform unifying device protocols behind a GraphQL API.

- Built frontend interfaces, SDKs, and internal tooling for a cloud IoT platform exposing a unified GraphQL API across multiple device protocols.
- Designed authentication, update, and upstream synchronization flows using TypeScript, React, MobX, and Apollo GraphQL.

Substrakt | Software Developer | Aug. 2018 – Jun. 2019
Digital product agency building bespoke publishing, arts, and cultural web platforms.

- Built bespoke WordPress features, search experiences, and internal tooling for design-led client websites.
- Modernized the JavaScript toolchain from Gulp to Webpack, improving development and deployment workflows.

D4 Software | Software Engineer | Jun. 2017 – Jul. 2018
Product engineering consultancy building client platforms, tools, and cloud infrastructure.

- Built client-facing features and internal tools across product and platform engagements using JavaScript and .NET.
- Maintained AWS infrastructure, serverless services, databases, and virtual machines while modernizing the frontend stack with Webpack, TypeScript, and Vue.js.

WRITING & SPEAKING

Writing: Multi-Agents; Enterprise Demands; Parallelising Development; Cosine CLI 1.16 Update
Topics: AI coding agents, developer workflows, enterprise deployment, secure agentic systems, multi-agent software engineering

EDUCATION

Birmingham City University | Bachelor's Degree, Computer Science with Year in Industry | 2015 – 2019

AREAS OF EXPERTISE

AI Developer Infrastructure: Go, TypeScript, Node.js, React, Next.js, GraphQL, CLI tooling, SDK-oriented APIs, IDE integrations, WebAssembly
Agent & Sandbox Systems: AI agents, browser use, memory systems, event-driven orchestration, tool execution, gVisor, Kubernetes, sovereign deployment
Reliability & Product Systems: AWS, observability, logging, tracing, VPC, on-prem, air-gapped environments, high-ownership product engineering